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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 03/23/2006 |
| **PI Name:** | Ciszak, Ewa M. Ph.D. | | |
| **Project Title:** | Crystal Structure of Human Pyruvate Dehydrogenase complex Facilitated by Microgravity | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | BIOTECHNOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | BIOTECHNOLOGY--Structural biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Huntsville | **State:** | AL |
| **Zip Code:** | 5812 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-03 |
| **Start Date:** | 09/01/2001 | **End Date:** | 11/30/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Patel, Mulchand S. ( SUNY at Buffalo )<br>Hong, Young S. ( BAE Analytical ) | | |
| **Grant/Contract No.:** | NAG8-1924 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | The proposed research falls in the category of the test of microgravity and the use of the International Space Station as a platform for research in the area of biotechnology and design of macromolecular materials. The research involves crystallization of component proteins of human pyruvate dehydrogenase complex using the terrestrial and low gravity environments followed by the analyses of X-ray diffraction data and structural determinations. These efforts should allow determination of benefit of microgravity for improving or not crystal quality for hPDC. Human pyruvate dehydrogenase complex (hPDC) is an intramitochondrial multienzyme complex consisting of multiple copies of several subunits including three enzyme activities E1 (pyruvate dehydrogenase), E2 (dihydrolipoamide acetyltransferase) and E3 (dihydrolipoamide dehydrogenase), required for the completion of the conversion of pyruvate to acetyl CoA, and E3BP (E3-binding protein) required for binding E3 component enzyme in this protein assembly. E1 catalyses the non-reversible removal of carbon dioxide from pyruvate; E2 forms acetyl CoA and E3 reduces NAD+ to NADH. Two | | |

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| **Task Description:** | additional enzyme activities are associated with the complex: a specific kinase which is capable of phosphorylating E1 at three serine residues and a loosely-associated specific phosphatase which reverses the phosphorylation The proportion of the PDC in its active dephosphorylated state is determined by a balance between the activity of the kinase and phosphatase. The activity of the kinase may be regulated in vivo by the relative concentrations of metabolic substrates such as pyruvate, lipoic acid covalently bound to E2, NAD+/NADH, CoA/acetylCoA as well as adenine diphosphate (ADP)/ATP. We propose to carry out X-ray crystallographic studies of component proteins of arguably the most challenging multienzyme assembly of the human body. Our research will focus on obtaining crystals of component proteins of the core of hPDC including E1, E2-binding fragment and E3, and determination of their structures. Crystallization conditions will also be tested in low gravity environment to demonstrate its value in improving crystal quality. We will also take care to construct validation protocols that should allow us to determine the impact of microgravity on the progress of our structural studies. Our research is designed to (1) validate the advantage of low-gravity as a tool for improving crystal quality and (2) achieve the structure of proteins of arguably the most challenging multienzyme assembly, human pyruvate dehydrogenase complex. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | N/A |
| **Task Progress:** | No progress report this reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 01/13/2005) |